

Cloud Computing Introduction and Foundations Concepts

Evolution towards Cloud Computing

Reference:

- [cam-san] Chapter 2

Cloud Computing - Definition

- **NIST, the US National Institute of Standards and Technology:** *“Cloud Computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort of service provider interactions”*
- **My Mom:** “The place where my phone saves my emails, contacts, pictures, so I don’t lose them”

Evolution of technology

- Cloud computing is not an abrupt innovation, instead, **it is the result of a series of developments that have taken place over the past few decades**
- The concept traces back to 1960, when McCarthy (the same who coined the term 'artificial intelligence') wrote:
 - *'computation may someday be organized as public utility', like water, electricity, gas, etc...*
- It took, however, more than 40 years for technology to develop and mature for cloud computing: only the composition of some key technologies has enabled cloud computing
- Several decades of research, especially in the domain of parallel and distributed computing, have paved the way

The origin – Mainframes

- Commercial usage of computing started around 1970s with mainframes
- Organizations acquired such hardware to automate basic data processing, e.g. payroll management
- Mainframes were large supercomputers installed in a room. They were so high priced that it *was hard for a company to have more than one*
- Multiple users were sharing the mainframe at the same time: the common approach was to implement a time-shared utilization
- Different users were accessing the mainframe from terminals

The origin – Mainframes

- Terminals were dumb terminals, i.e. were only capable of transferring input-output data to/from the mainframe (the server)
- This centralized processing environment was a bottleneck
- Users had always to wait for a long time in queue



PC Revolution

- At the end of the 70s less expensive computers came into the market
- Those Personal Computers (PCs) provided some processing and storage capabilities to run applications
- The need to maintain expensive mainframes systems soon became less important

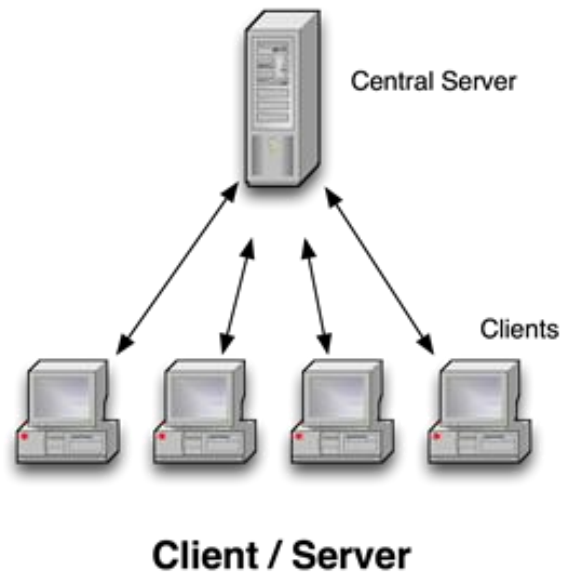


Network of PCs

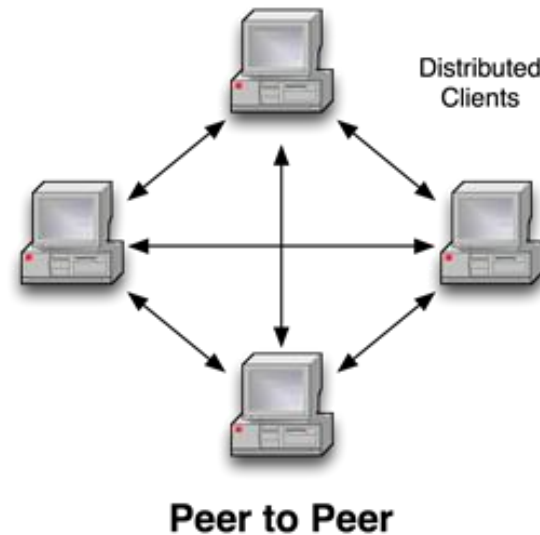
- Big organizations installed multiple PCs in different offices
- Each one, however, was functioning independently
- Network communication technologies were introduced to allow communication between PCs for data transfer
- LAN: Local Area Network (LAN) and Wide Area Network (WAN) were introduced at that time: *computer in the same office or placed at distance could communicate*
- Over time communication technologies improved, increasing the bandwidth

Distributed applications

- Client/Server computing: we have a server (offering some service/information) and a client



- Peer to peer computing: each and every computer can play the role of a client as well as a server



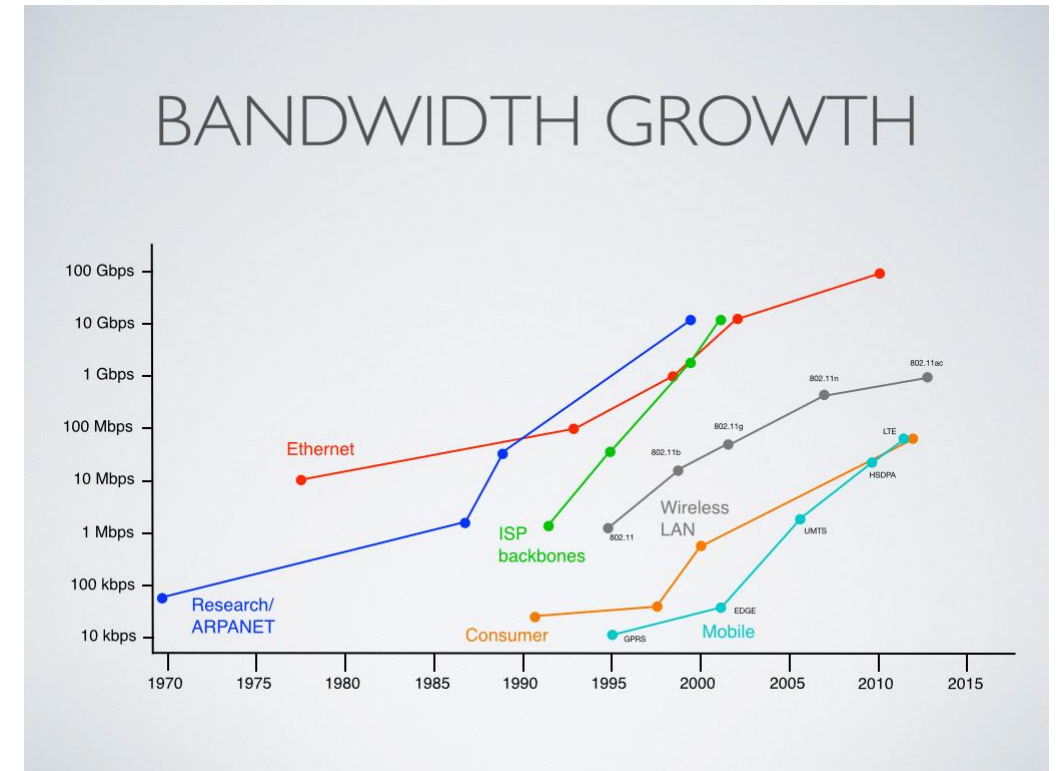
Interaction was limited to exchange of small data due to network limitations

Parallel processing

- In early 1980s it was believed that computer performance could only be improved by producing faster and more powerful processors
- ***Parallel processing*** changed this idea: multiple processors (installed on the same PC or on different PCs) working together to solve a single (big) task that could not be executed on a single one
- The task must be designed to be parallel, i.e. composed by a set of subtasks that could be executed in parallel

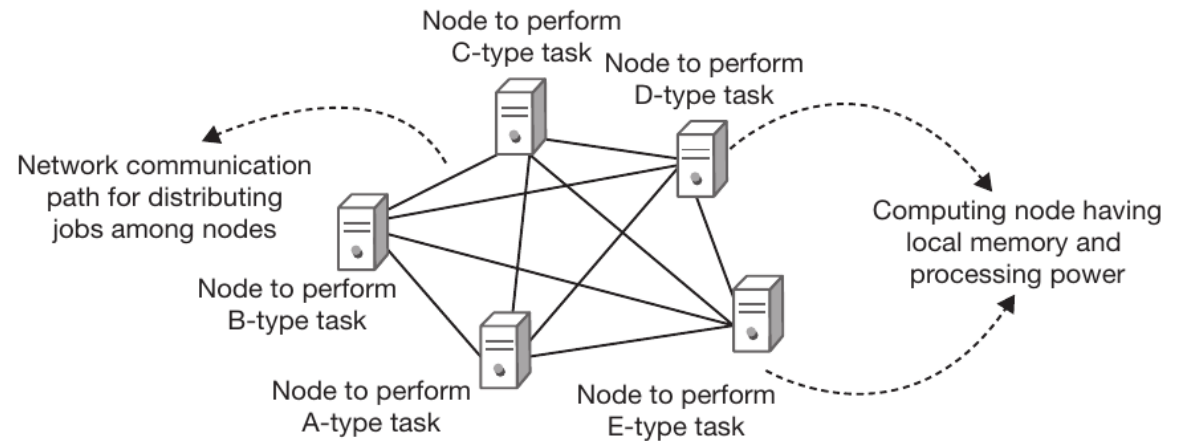
Faster Network Communication

- During 1980s advancements in network communication improved significantly data transfer rates (up to 100 Mbps for LANs and up to 64 Kbps in WANs)
- Such advancements enabled ***distributed computing systems***: a collection of processors interconnected via communication network to execute a complex task, impossible to be executed on a single PC



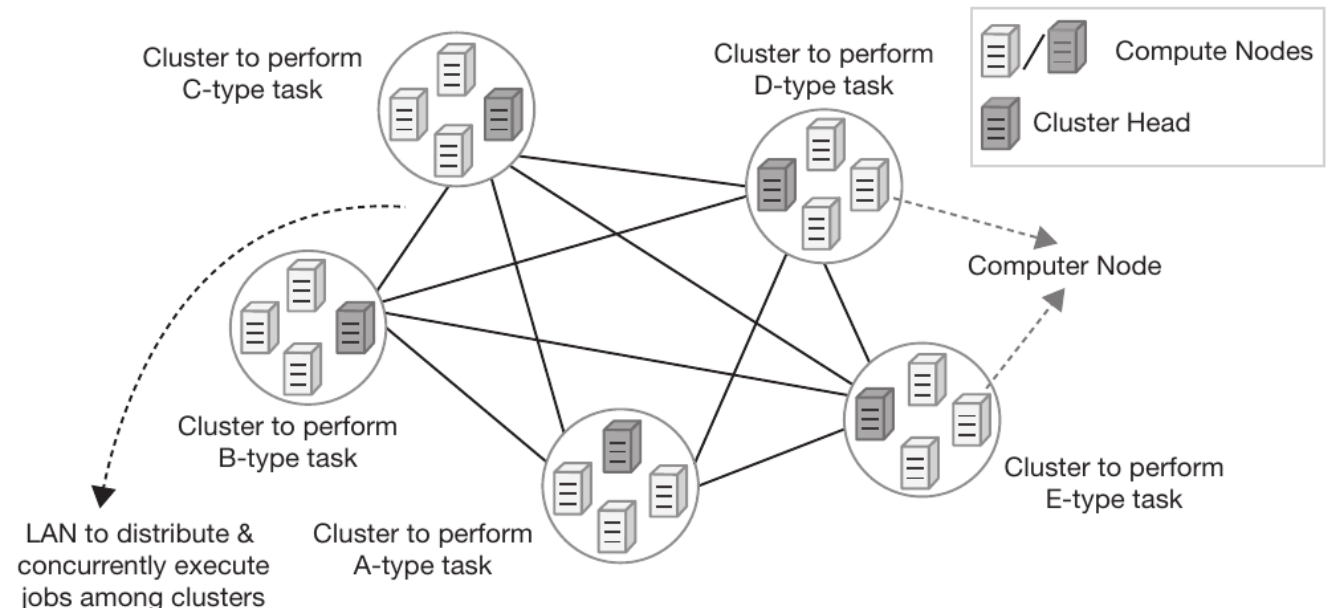
Distributed computing systems

- Applications are programmed to be executed in a distributed manner as a composition of different tasks
- Each system has its own local memory and processing power
- Nodes communicate to exchange data or results



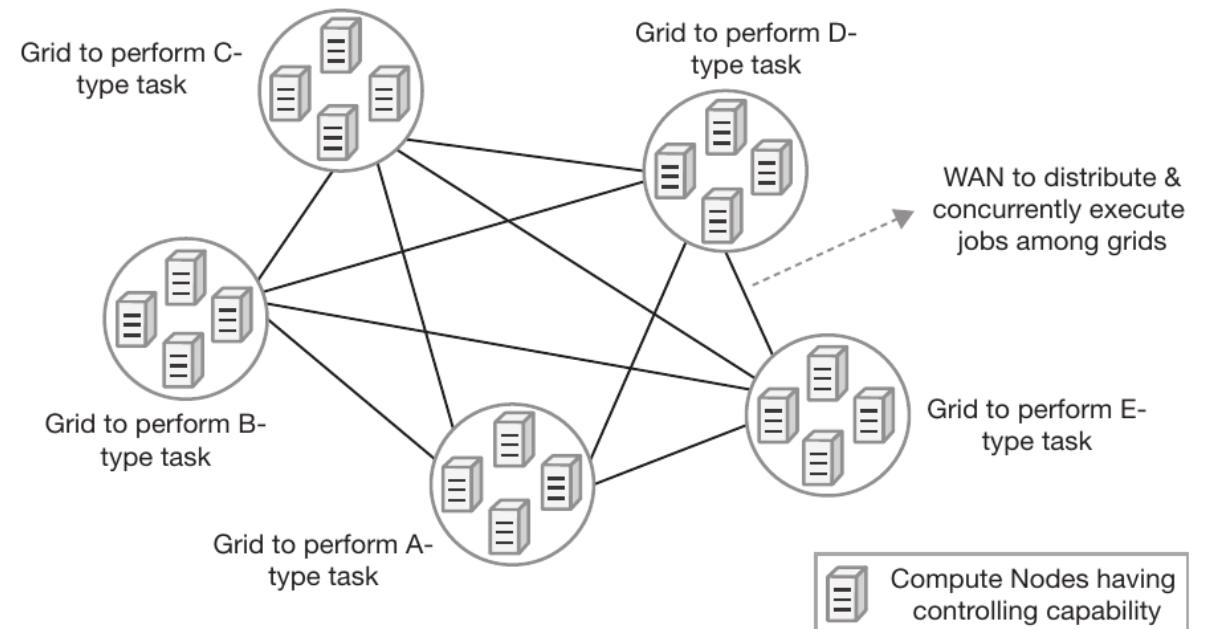
Cluster computing

- The next natural step was the creation of clusters: groups of multiple nodes (computers) all connected to the same LAN to perform the same (or similar) tasks
- Computers are clustered together for a more efficient computing (less communication overhead) and to improve reliability.
- Pools of homogeneous computing systems were deployed to have more processing power
- Among the computers of a cluster, one node was selected as cluster head to control the cluster and distribute the tasks



Grid Computing

- The cluster head represented a single point of failure
- To mitigate this risk in early 90s a new architecture was introduced, **grid computing**: *a pool of resources in which the control functionalities were implemented in a decentralized manner (not in the cluster head)*
- Computing nodes could be deployed in the same network or could be deployed in different areas (even under different administrative domains)



Not enough

- Grid computing provided many advantages: *a scalable architecture that could offer high performance computing for complex tasks*
- *It had many drawbacks:*
 - Real-time scaling was not possible: more computing power/storage -> more nodes -> change in the architecture
 - The system was not fault tolerant: the failure of a node results in the failure of a task
 - Heterogeneous hardware required code adaptation
- Starting from the late 90s a new set of technologies matured, paving the way for cloud computing

Hardware Virtualization

- The heterogeneous natures of computing systems architectures posed a challenge to application (software) portability
- The execution of software on different architecture without changes required to decouple software systems from the underlying physical resources of computers
- This would allow to deal with hardware diversity but also to achieve real time scalability
- For this reason, virtualization techniques were defined: a layer of software over the hardware system that could simulate the whole physical system environment



We will see that
extensively, don't worry!

Web Based Technologies

- As computing technologies were evolving rapidly, users and business were becoming more dependent on computer systems
- In addition to enabling grid computing to empower the development of large computing systems, faster networks allowed users from different geographical locations to collaborate in almost real time
- The World Wide Web arose as the killer application to spread information and enable collaboration
- Web 2.0 appeared after 2002 as an evolution of Web 1.0, in which user generated content was central

Service Oriented Architecture

- As the number of IT systems and their size increased, a new model for software development was required
- Service Oriented Architecture (SOA) is a development method in which applications are developed by leveraging software components (software services) interacting each other
- Each service is an independent entity that communicate with the others via messages
- Systems implemented using SOA remain flexible to changes



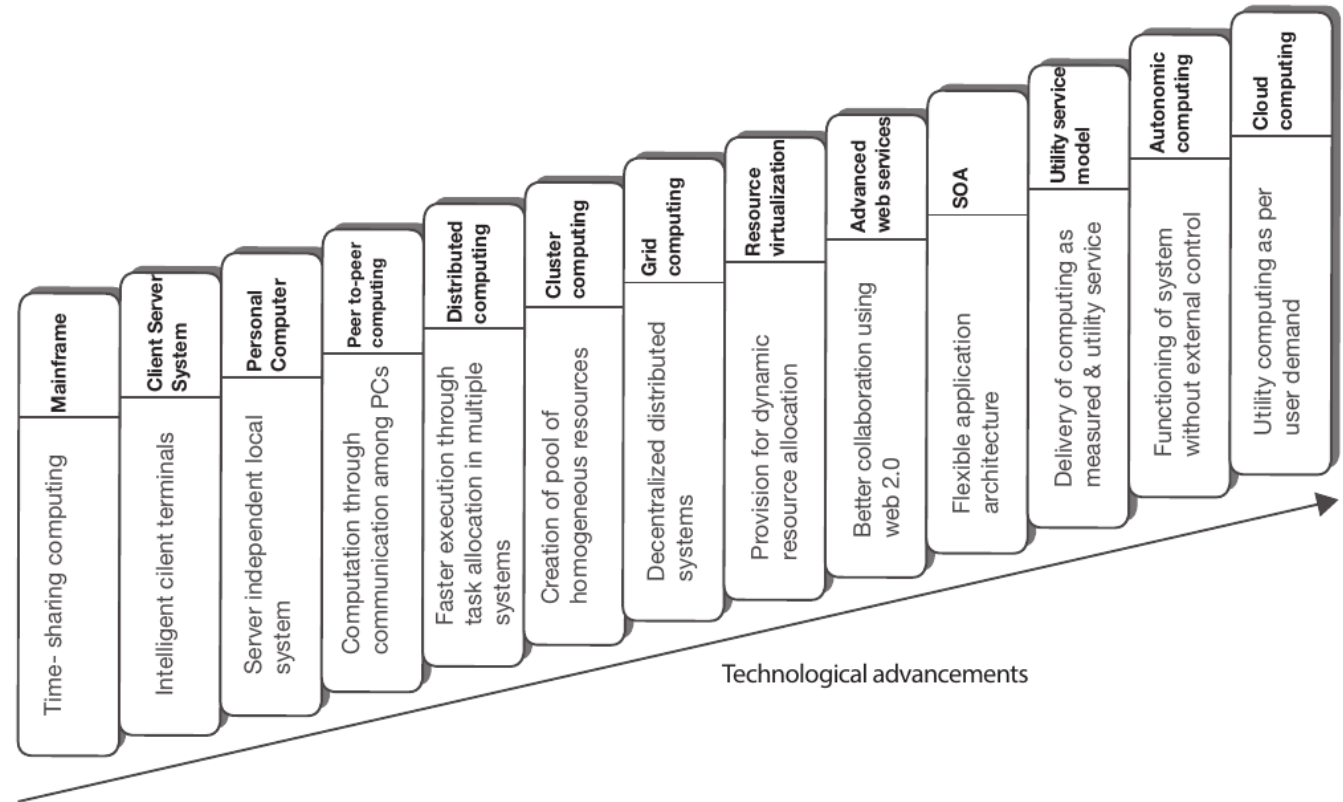
We will see that,
don't worry!

Utility Computing Model

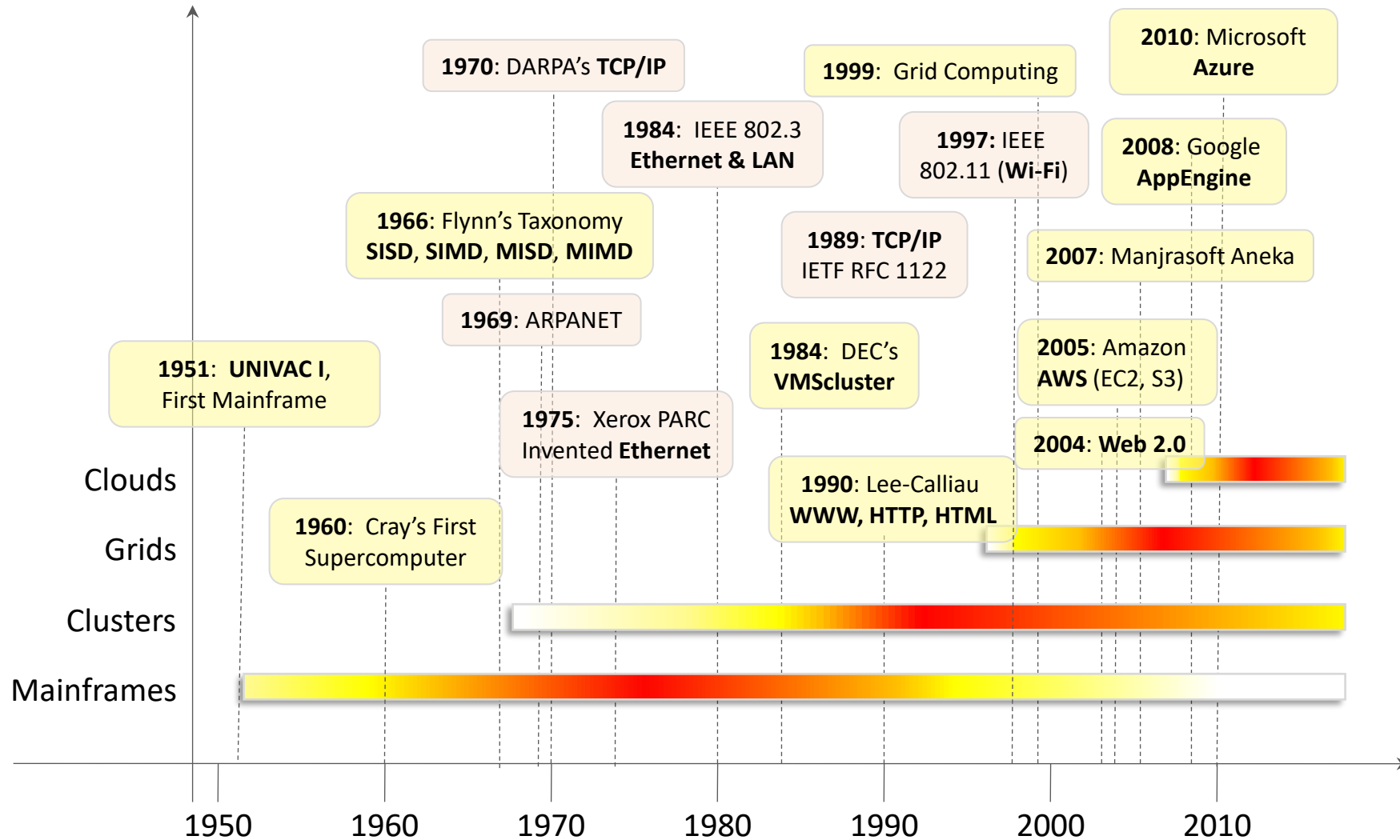
- Computing era reached a point in which we had:
 - Scalable computing infrastructure made of heterogeneous resources
 - Distributed computing environments empowered by high bandwidth networks
 - Collaborative environment across the world via web services
 - Flexible applications via SOA
- Utility Computing: delivering of computing resources as utility, following a pay per use model and on demand
- Crucial **enabler** for this model was hardware virtualization

Cloud Computing ... eventually

- Utility computing become cloud computing
- Each one was essential, from networking advancements to novel software development methodologies



Timeline



Cloud computing today

- The first commercial initiative based on cloud computing concepts was from salesforce.com in 1999
- The first large scale cloud computing service was commercialized by Amazon in 2006 that launched its Elastic Cloud Computing (EC2) and Simple Storage Service (S3)
- Soon after other big players entered in the market with similar services, e.g. Microsoft in 2009
- The term Cloud computing appeared with its present meaning for the first time in 2006, used by Google CEO Eric Schmidt in a conference

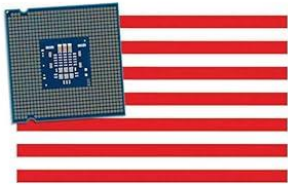
Cloud computing in numbers

- The total expenditure worldwide reached a figure of \$210 billion at the end of 2020
- Demand for cloud services grows by 18% in 2019
- 90% of companies use some type of cloud service
- 80% of enterprises use Amazon Web Services as their primary cloud platform
- There were 3.6 billion cloud users in 2018
- The positive impact of cloud technology is almost instantaneous. 80% of companies report operation improvements within the first few months of adopting the tech

NEW YORK TIMES BESTSELLER

"Pulse quickening. If any book can make general audiences grok the Silicon Age—and finally recognize how it rivals the Atomic Age for drama and import—*Chip War* is it." —*THE NEW YORK TIMES*

CHIP WAR



THE FIGHT FOR THE WORLD'S
MOST CRITICAL TECHNOLOGY

CHRIS MILLER

Good book for IT history
CHIP WAR by Chris Miller